

Homework #2 (ChE 451)

The semi-trucks used in the US trucking industry tend to be very large and travel long distances. As a result, semi-trucks usually use diesel fuel rather than gasoline because it is more efficient. Diesel fuels have about 10% to 15% more energy than gasoline, therefore allowing diesel vehicles to go 20% to 35% farther than a vehicle using gasoline [1]. Even with the use of diesel fuel, semi-trucks are very big and require more energy just to accelerate and keep going for the long distances required for them to transport cargo. With that said, new ways to increase fuel efficiency have been explored such as the trailer tail, an aerodynamic design.

The trailer tail was designed to increase fuel efficiency by reducing drag. Drag is created by the airflow when a truck is moving at a high and constant speed [2]. The airflow works against the front of the truck creating a high-pressure resistance. Drag is also created by the air flowing underneath the truck. With this drag, a low-pressure vacuum is created at the back of the truck, pulling it back. Without any aerodynamic designs, such as the trailer tail, drag accounts for 65% of fuel consumption by semi-trucks. By adding the trailer tail, the drag is reduced and enhances the fuel efficiency by over 5% [3].

In order to evaluate the true efficiency of trailer tails several points need to be taken into consideration. Wind patterns differ for trucks traveling from North to South compared to trucks traveling East to West [4]. With that being said, the economic benefits will vary depending on where the trucks are traveling. It is also difficult to accurately calculate how much the trailer tail is contributing to the fuel efficiency of the truck. To increase the efficiency, truck

fleets usually install several aerodynamic designs such as dry van trailers, side skirts, and “boat tails”. The National American Council for Freight Efficiency (NACFE) reports that because fleets utilize more than one kind of aerodynamic design, it makes it harder for companies to decide if investing in trailer tails will compensate its price in a reasonable time frame [4]. Based off that, another important point that may justify installing a trailer tail is considering diesel fuel prices. A gallon of diesel fuel is \$3.17 [5] and allows a truck to travel 5.7 miles [6] and, according to the Federal Highway Administration, trucks may travel up to 100,000 miles a year, if not more [7]. This will equate to about \$70,000 spent on fuel a year for a single semi-truck. With 5% of that \$70,000 saved a year, a company can save upwards of \$3,500 a year per truck, after the first-year installation cost of \$2,220 for a single trailer tail [2].

Other positives that are associated with trailer tails are that they stabilize the truck in windy conditions [3]. This will create a safer trip for both the truck drivers and the people driving vehicles on the same road. It is safer because the more stable the truck is, the less of a need for lane corrections and therefore reducing fatigue experience by the driver [4]. Another positive to installing trailer tails are they can reduce the amount of carbon dioxide put into the atmosphere by 4.9 metric tons, which is equivalent to taking a car off the road for a year [2].

Overall, from an economic stance, trailer tails have been seen to improve fuel efficiency and even compensate for their hefty initial cost as discussed above. They also have several positive characteristics associated with them such as increasing safety on the road and reducing the truck industry’s carbon footprint. While there may be some setbacks to trailer tails, such as truck drivers forgetting to fold the flaps before backing up [2], training drivers to add this step into their standardized operating procedures will prevent the problem, if not eliminate it.

References

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